ECE 461/661 Spring 2026 Mini-Exam 1

Introduction to Machine Learning for Engineers

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Mon, February 9, 2025, 9:00am - 10:00am PT/12:00pm - 1:00pm ET/7:00pm - 8:00pm CAT

Instructions

a. If a problem asks you which of its choices is TRUE, you should treat choices that may be either true or false as FALSE.

b. **Unless otherwise stated, only one option is correct in each multiple-choice question.** No partial credit will be given for multiple-choice or true/false questions.

c. For descriptive questions, make sure to explain your answers and reasoning. We will give partial credit for wrong answers if portions of your reasoning are correct. Conversely, correct answers accompanied by incomplete or incorrect explanations may not receive full credit.

d. You are allowed one **physical, single-sided** US-letter or A4 sized cheat sheet. No other notes or material (aside from blank pieces of scratch paper) may be used.

e. You may only use a pen/pencil, eraser, and scratch paper. The backside of each sheet in the exam can also be used as scratch paper. If you do not wish for us to grade your scratch work, please clearly indicate which parts of your work we should ignore.

f. Calculators are permitted but not necessary. If you choose to use a calculator, it must be a standalone one. No other electronic devices such as phones, tablets or laptops can be used during the exam.

g. If you would like to ask a clarification question during the exam, raise your hand and an instructor or TA will come over. Note that we will not help you answer the questions but can give clarifications.

Problem	**Type**	**Points**
1-5	True/False	7
6-8	Multiple Choice	6
9	Descriptive	4
10	Descriptive	3
11	Honor Pledge	0
Total		20

1 True or False (7 points)

Problem 1: *[1 points]* A data scientist trained a regression model for house price prediction. After training the model, the ground-truth price labels were lost due to a storage failure. The data scientist cannot estimate the model's variance without these labels.

◯ True

◯ False

Solution: True. Model variance cannot be estimated without ground-truth labels because we need the labels to retrain on different resampled versions of the training data.

Problem 2: *[3 points]* A hospital trains a linear regression model to predict patient outcomes using tabular features. Initially, the dataset contains 500 patients and 5 features per patient, where the features have very different magnitudes.

a. The clinicians compare the results obtained using ridge regression and linear regression and find that ridge regression has a higher training error. They increase the regularization parameter λ, which will improve the training error, and thus the generalizability, of the ridge regression model.

◯ True

◯ False

Solution: False. Training error does not measure generalizability, and increasing λ may improve or worsen test performance, so λ should be tuned on validation data. Moreover, if the goal is to reduce training error, λ should be decreased to make ridge regression closer to linear regression.

b. The clinicians set the ridge regularization parameter to a negative value, $\lambda < 0$. This regularization method will still control model complexity, and there still exists a unique global minimizer for any dataset.

◯ True

◯ False

Solution: False. The Hessian is $X^\top X + \lambda I$, which may fail to be positive semi-definite when $\lambda < 0$. The objective is no longer guaranteed to be convex and can even be unbounded below, so a minimizer may not exist.

c. The clinicians add another 500 patients to the dataset and collect 5 additional features for every patient. Batch Gradient Descent (BGD) would run faster than either Stochastic Gradient Descent (SGD) or the closed-form solution, assuming that BGD converges in 20 rounds and SGD converges in 10,000 rounds on this dataset.

◯ True

◯ False

Solution: False. After the updates, $N = 1000$ and $D = 10$. The closed-form solution costs $O(ND^2 + D^3) \approx 1000 \cdot 10^2 + 10^3 = 101,000$. BGD costs $O(ND)$ per round, so 20 rounds cost $\approx 20 \cdot 1000 \cdot 10 = 200,000$. SGD costs $O(D)$ per round, so 10,000 rounds cost $\approx 10,000 \cdot 10 = 100,000$. Thus, BGD is comparably slower than the other two methods.

Problem 3: *[1 points]* In ridge regression with regularization parameter $\lambda > 0$, as $\lambda \to \infty$, both the bias and the variance of the estimator go to zero.

◯ True

◯ False

Solution: False. As $\lambda \to \infty$, ridge regression shrinks all coefficients toward zero. This makes the model increasingly rigid, which reduces variance but increases bias, since the predictions systematically deviate from the true underlying function. Therefore, both bias and variance do not go to zero simultaneously.

Problem 4: *[1 points]* In the probabilistic interpretation of ridge regression with feature vector $\mathbf{x}$, parameter vector $\mathbf{w}$ and target variable y, we perform maximum a posteriori (MAP) estimation using a Beta prior on the parameter vector $\mathbf{w}$.

◯ True

◯ False

Solution: False. We use a Gaussian prior on $\mathbf{w}$, not a Beta prior.

Problem 5: *[1 points]* You are given a design matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$. First, you apply the nonlinear basis function $\phi(x) = x^2$ elementwise to $\mathbf{X}$ to obtain $\mathbf{\Phi}$. Then you compute the ridge regression solution

$$\mathbf{w}_{\text{ridge}} = (\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda \mathbf{I})^{-1} \mathbf{\Phi}^\top \mathbf{y},$$

where $\lambda > 0$ is a given regularization parameter and $\mathbf{y}$ denotes the target vector. Recall that the ordinary least squares (OLS) solution $\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ has computational complexity $O(ND^2 + D^3)$.
Even though ridge regression requires the additional operation of adding $\lambda \mathbf{I}$ to $\mathbf{\Phi}^\top \mathbf{\Phi}$ before inversion, and we apply a nonlinear transformation $\phi(x) = x^2$ to every element of $\mathbf{X}$ to obtain $\mathbf{\Phi}$, the dominant computational complexity of computing $\mathbf{w}_{\text{ridge}}$ remains $O(ND^2 + D^3)$.

◯ True

◯ False

Solution: True.
Although ridge regression requires adding $\lambda \mathbf{I}$ (costing $O(D^2)$) and applying ϕ requires $O(ND)$ operations, both are dominated by the existing $O(ND^2 + D^3)$ complexity. The matrix multiplication $\mathbf{\Phi}^\top \mathbf{\Phi}$ still costs $O(ND^2)$, and the inversion still costs $O(D^3)$. Therefore, the dominant complexity remains $O(ND^2 + D^3)$.

2 Multiple Choice (6 points)

Problem 6: *[2 points]* Suppose you compute the singular value decomposition (SVD) of a 4×3 matrix $A \in \mathbb{R}^{4\times 3}$, where A has rank 2. Let σ_1, σ_2 denote the nonzero singular values of A with $\sigma_1 > \sigma_2$; let $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$ denote the (right) eigenvectors of A; and let $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ denote the eigenvectors of $A^T A$. Assume the eigenvectors are ordered in decreasing order of their corresponding eigenvalues.
Let $A = U\Sigma V^T$ denote the SVD of A. What are Σ and V?

◯ $\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $V = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \end{bmatrix}$

◯ $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $V = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \end{bmatrix}$

◯ $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $V = \begin{bmatrix} \boldsymbol{w}_1 & \boldsymbol{w}_2 & \boldsymbol{w}_3 \end{bmatrix}$

◯ No SVD exists since A is not full rank.

Solution: A.

Problem 7: *[2 points]* You observe $n_H = 12$ heads and $n_T = 8$ tails from 20 independent flips of a coin, where n_H denotes the number of observed heads and n_T denotes the number of observed tails. You wish to compute the maximum likelihood estimate (MLE) and maximum a posteriori estimate (MAP) of the probability of observing "heads." To do so, you use a Beta prior with parameters (α, β) and decide to set $\alpha = \beta$. How do the MAP and MLE estimates change as you increase $\alpha = \beta$?

◯ The MLE estimate approaches $\frac{\alpha}{\alpha+\beta} = \frac{1}{2}$.

◯ The MAP and MLE estimates will become more similar.

◯ The MAP estimate approaches $\frac{1}{\alpha+\beta} = 0$.

◯ The MAP estimate approaches $\frac{\alpha}{\alpha+\beta} = \frac{1}{2}$.

Solution: D. As $\alpha = \beta$ increases, the MLE estimate of $\frac{n_H}{n_H+n_T} = \frac{3}{5}$ will not change and the MAP estimate will converge to $\frac{\alpha}{\alpha+\beta} = \frac{1}{2}$.

Problem 8: *[2 points]* For Stochastic Gradient Descent (SGD) and full-batch gradient descent (GD), which of the following statements are true? **More than one option can be true.**

☐ In each iteration of SGD, one sample is drawn uniformly at random without replacement to ensure that the gradient is an unbiased estimate of the gradient used in full-batch GD.

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☐ SGD can get stuck in local minima, whereas GD is guaranteed to reach the global minimum of a non-convex objective function.

☐ GD can get stuck in local minima, whereas SGD is guaranteed to reach the global minimum of a non-convex objective function.

Solution: B

- A is False; because without replacement sampling will lead to biased stochastic gradients
- B is True; because with replacement sampling will lead to unbiased stochastic gradients
- C and D are both False; SGD and GD can both get stuck in local minima

3 Descriptive (7 pts)

Problem 9: *[4 points]*
A reward system works as follows. Each user makes attempts until they obtain the rare reward, at which point they **stop immediately**. The success probability increases by a known constant $\delta > 0$ after the first attempt. In this problem, assume the system is designed so that the third attempt is guaranteed to succeed, meaning

$$p_1 = p, \qquad p_2 = p + \delta, \qquad p_3 = 1,$$

where p is unknown and we assume $0 < p < 1 - \delta$. Here p_m for $m = 1, 2, 3$ denotes the probability of receiving the reward in the mth attempt, given that the mth attempt is made.
Let $X \in \{1, 2, 3\}$ be the attempt on which a user first succeeds. We observe n independent users. Let n_1, n_2, n_3 be the numbers of users with $X = 1, 2, 3$, with $n_1 + n_2 + n_3 = n$.

(a) Write down $\Pr(X = 1)$, $\Pr(X = 2)$, and $\Pr(X = 3)$ in terms of p and δ.

Solution: Since users stop at first success,

$$\Pr(X = 1) = p, \qquad \Pr(X = 2) = (1 - p)(p + \delta), \qquad \Pr(X = 3) = (1 - p)(1 - (p + \delta))$$

(b) Write down the log-likelihood function $\ell(p)$, and derive the first-order condition that the MLE must satisfy, namely $\ell'(p) = 0$ (no need to solve it).

Solution: The log-likelihood is

$$\ell(p) = n_1 \log p + n_2\big(\log(1 - p) + \log(p + \delta)\big) + n_3\big(\log(1 - p) + \log(1 - p - \delta)\big)$$

Differentiating and setting $\ell'(p) = 0$ gives the first-order condition

$$\ell'(p) = \frac{n_1}{p} - \frac{n_2 + n_3}{1 - p} + \frac{n_2}{p + \delta} - \frac{n_3}{1 - p - \delta} = 0$$

(c) Suppose p has a Beta(α, β) prior distribution with density

$$\pi(p) \propto p^{\alpha - 1}(1 - p)^{\beta - 1}, \qquad 0 < p < 1$$

Write down the log-posterior, and derive the first-order condition that the MAP estimator must satisfy (no need to solve it).

Solution: The log-posterior is

$$\ell(p) + (\alpha - 1)\log p + (\beta - 1)\log(1 - p),$$

where $\ell(p)$ is the log-likelihood from part (b). Differentiating gives the first-order condition

$$\ell'(p) + \frac{\alpha - 1}{p} - \frac{\beta - 1}{1 - p} = 0$$

Substituting the expression for $\ell'(p)$ from part (b), the MAP estimator satisfies

$$\frac{n_1 + \alpha - 1}{p} - \frac{n_2 + n_3 + \beta - 1}{1 - p} + \frac{n_2}{p + \delta} - \frac{n_3}{1 - p - \delta} = 0$$

WRITE YOUR ANSWER TO PROBLEM 9 BELOW

CONTINUE YOUR ANSWER TO PROBLEM 9, IF NEEDED

Problem 10: *[3 points]* Consider the following dataset with two data points and two input features:

$$\left(\begin{bmatrix}1\\-1\end{bmatrix}, 1\right) \quad \left(\begin{bmatrix}-1\\1\end{bmatrix}, 1\right)$$

Here each data point is expressed as $\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}, y\right)$, where x_1 and x_2 are the two features and y is the label.

(a) Suppose that you use linear regression to learn parameters w_1, w_2 so as to minimize the residual sum-of-squares (RSS)

$$(w_1(1) + w_2(-1) - 1)^2 + (w_1(-1) + w_2(1) - 1)^2 .$$

Find the unique closed-form solution (w_1^*, w_2^*), or explain why such a solution does not exist.

Solution: We form the data matrix $X = \begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix}$. We can see by inspection that this matrix has rank 1 and thus will not yield a unique solution. Alternatively, we can compute $X^T X = \begin{bmatrix}2 & -2\\-2 & 2\end{bmatrix}$, which is not invertible. Thus, a closed-form solution does not exist.

(b) Suppose that we use gradient descent to find a solution to our linear regression problem from part (a). We use suitably chosen learning rates so that the gradient descent algorithm converges. Will the resulting solution lead to a RSS of 0? In 1-2 sentences, explain why or why not.

Solution: No it will not, since $w_1(1) + w_2(-1) = -(w_1(-1) + w_2(1))$. Thus, the predictions of our two data points will be negatives of each other, but their labels are the same, so they cannot achieve a RSS of 0.

(c) Now suppose that we include a bias term in our linear model, i.e., we wish to solve for (w_0, w_1, w_2) to minimize the residual sum-of-squares between the labels y of our two data points and $w_0 + w_1 x_1 + w_2 x_2$. Suppose we again use gradient descent to find a solution, with suitably chosen learning rates so that the gradient descent algorithm converges. Will the resulting solution lead to a RSS of 0? In 1-2 sentences, explain why or why not.

Solution: Yes, because we can achieve 0 RSS (thus minimizing it) at $w_0 = 1, w_1 = w_2 = \alpha$ for any $\alpha \in \mathbb{R}$. Although this solution is not unique, gradient descent will converge to a point within this solution set.

WRITE YOUR ANSWER TO PROBLEM 10 BELOW

CONTINUE YOUR ANSWER TO PROBLEM 10, IF NEEDED

4 Honor Pledge

Problem 11: *[0 points]* To affirm that you did not cheat on the exam, please write out the below statement. Sign your name beneath it. **Failure to do so will be taken as a sign that you have cheated on the exam.**

I pledge my honor that I neither gave nor received unauthorized assistance on this examination.